Physics
Higher level
Paper 1A and Paper 2
B.4 Thermodynamics

8 November 2026

W4ss Academy practice test 3 | **Topic** B.4 Thermodynamics | **Ref.** W4SS–PHY–B4–03

Student name

30 minutes [Paper 1A and Paper 2]

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- **Paper 1A:** for each question, choose the answer you consider to be the best and indicate your choice on the answer sheet provided.
- **Paper 2:** answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **physics data booklet** is required for this paper.
- The maximum mark for paper 1A is **[3 marks]**.
- The maximum mark for paper 2 is **[17 marks]**.
- The maximum mark for this examination paper is **[20 marks]**.

Paper 1A

1. Which statement is correct about the entropy of a non-isolated system?

A. It always increases

B. It always decreases

C. It can only increase if the entropy of the surroundings decreases

D. It can only decrease if the entropy of the surroundings increases

2. For a thermodynamic process, the entropy of the universe

A. always increases during the process

B. depends only on energy transferred during the process

C. is zero during the process

D. never decreases during the process

3. A working refrigerator with the door open is placed in a sealed room.

The entropy of the room

A. is zero.

B. decreases.

C. remains unchanged.

D. increases.

Paper 1A — answer sheet

Student name _____ Date _____

1.	A	B	C	D
2.	A	B	C	D
3.	A	B	C	D

Shade **one** circle per question. If you change your answer, cross out the wrong circle clearly.

Paper 2

Answer **all** questions. Answers must be written within the answer boxes provided.

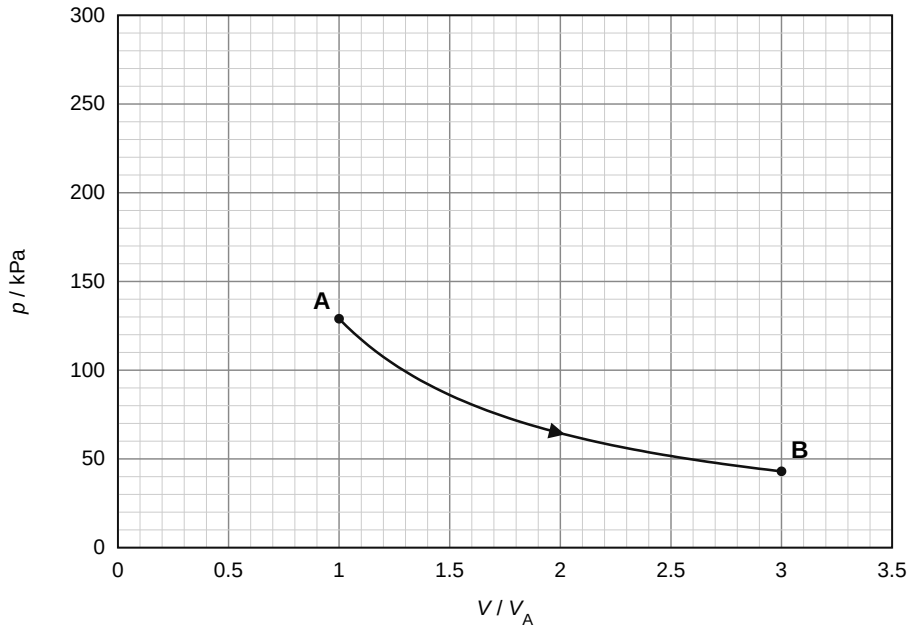
1. A fixed quantity of an ideal monatomic gas is trapped in a cylinder. At A the pressure of the gas is 129 kPa and its temperature is 27 °C. The gas undergoes the following cycle:

A → B: isothermal expansion until the volume is three times the volume at A

B → C: adiabatic compression back to the volume at A

C → A: cooling at constant volume back to A

The *pV* diagram shows the process AB.



(a) Calculate the pressure of the gas at B. [2]

(This question continues on the following page)

(Question 1 continued)

(b) Sketch, on the *pV* diagram, the remaining two processes BC and CA that the gas undergoes. [2]

(c) Show that the temperature of the gas at C is approximately 350 °C. [2]

(d) Explain why the change of entropy for the gas during the process BC is equal to zero. [1]

(e) Explain why the work done by the gas during the isothermal expansion AB is less than the work done on the gas during the adiabatic compression BC. [1]

(This question continues on the following page)

(Question 1 continued)

(f) The quantity of trapped gas is 53.2 mol. Calculate the thermal energy removed from the gas during process CA. [2]

2. An isolated system consists of six distinguishable particles that share six quanta of energy. Each particle can have 0, 1, 2, … quanta.

In configuration A, one particle has all six quanta and the other five particles have none.

In configuration B, three particles have no energy and the other three particles have 1, 2 and 3 quanta.

(a) State and explain the number of microstates of the system in configuration A. [2]

(b) Configuration B has 120 microstates. Calculate the entropy difference between configurations B and A. State the answer in terms of k_B . [2]

(This question continues on the following page)

(Question 2 continued)

(c) The system is initially in configuration A. Comment, with reference to the second law of thermodynamics and your answer in (b), on the likely evolution of the system. [3]

Sources of the questions

This practice test was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Physics Questionbank (first assessment 2025), topic B.4 Thermodynamics. The introductions of the Paper 2 questions are missing from the source: W4ss Academy wrote them using only the data of the official markscheme, so that the markscheme still applies. The questions remain the property of the International Baccalaureate Organization and must not be published or sold.

Paper 1A — question 1	EXE.1A.HL.TZ0.20
Paper 1A — question 2	EXE.1A.HL.TZ0.24
Paper 1A — question 3	SPM.1A.HL.TZ0.12
Paper 2 — question 1	23M.2.HL.TZ2.7 — introduction written by W4ss Academy from the markscheme
Paper 2 — question 2	EXE.2.HL.TZ0.12 (parts (b), (c), (d)) — introduction written by W4ss Academy from the markscheme